Notes_25.07: Types of Naive Bayes Classifiers

Python 3.10+

Jupyter Notebook

PDF notes.pdf

🌐 **Language: English** | हिंदी / Hinglish | ⬅️ Lecture 25 Overview

📌 Executive Summary

GaussianNB for continuous features, MultinomialNB for discrete counts, and BernoulliNB for binary indicators.

🎯 Mathematical Formulations

Gaussian Naive Bayes Likelihood (Normal PDF):

$$P(x_j \mid y) = \frac{1}{\sqrt{2\pi\sigma_{j,y}^2}} \exp\left(-\frac{(x_j - \mu_{j,y})^2}{2\sigma_{j,y}^2}\right)$$

Multinomial Likelihood:

$$P(\mathbf{x} \mid y) = \frac{(\sum x_i)!}{\prod x_i!} \prod p_{i,y}^{x_i}$$

💡 Conceptual Principles & Architecture

Kaunsa Model Kab Lagayein?

1. **GaussianNB:** Jab features continuous real numbers hon (jaise Age, Blood Pressure, Temperature). Yeh assume karta hai ki har feature class ke andar

Gaussian/Normal distribution follow karta hai.

2. **MultinomialNB:** Text classification jahan word frequency count (Bag of Words / TF-IDF) hota hai (jaise text documents, sentiment analysis).
3. **BernoulliNB:** Jab features binary yes/no (1 ya 0) indicators hon (jaise word present hai ya absent).

💻 Python Production Implementation

```
from sklearn.naive_bayes import GaussianNB, MultinomialNB, BernoulliNB
import numpy as np

# Continuous features -> GaussianNB
X_cont = np.array([[1.2, 5.4], [2.1, 4.8], [5.6, 1.2], [6.2, 0.9]])
y = [0, 0, 1, 1]
gnb = GaussianNB().fit(X_cont, y)
print("GaussianNB Classes:", gnb.classes_)
```